

Diabetes Prediction Using Machine Learning

Machine Learning Project Report

Author: Shonu
B.Tech (AI & Data Science)

Overview

This project predicts whether a person has diabetes using Logistic Regression. It includes data preprocessing, model training, testing, and evaluation.

Tech Stack

- Python
- Jupyter Notebook
- Pandas
- Scikit-Learn

Dataset Description

The Pima Indians Diabetes Dataset contains medical diagnostic reports of 768 individuals. Features include Glucose, BMI, BloodPressure, Insulin, and Age. Target: 0 = Non-diabetic, 1 = Diabetic.

Project Structure

- diabetes_prediction.ipynb
- diabetes.csv
- project_report.txt
- README.md

Model Used

Logistic Regression is used for binary classification because it is efficient, interpretable, and widely used in medical ML applications.

Results

Final model accuracy: **74.67%**. This is within the expected performance range for this dataset.

Future Enhancements

- Add RandomForest, SVM, Naive Bayes, XGBoost
- Add visualizations and correlation heatmaps
- Hyperparameter tuning
- Deploy using Streamlit or Flask

Author

Shonu
B.Tech (AI & DS)

India